Finite propagation in fourth-order aggregation–diffusion

Kaiwen Jin*

Abstract

We prove finite propagation for a fourth-order aggregation–diffusion equation with linear mobility and quadratic attraction on $\mathbb{R}^d$. For every $d \geq 2$, every positive attraction coefficient and every compactly supported nonnegative H^1 probability density, there is one global weak solution whose support remains in a bounded ball on each finite time interval. The solution has the same $L^\infty H^1 \cap L^2 H^2$ regularity and integrated weak formulation as the variational existence theory. A localized power-entropy estimate is proved on the same minimizing movements by a bounded-derivative auxiliary flow. The resulting superlinear inequality for the exterior spacetime tail forces that tail to vanish at a finite radius, in every finite dimension.

1 Introduction and main theorem

We study the degenerate fourth-order equation

$$\partial_t \rho = -\operatorname{div}(\rho \nabla \Delta \rho) - \chi \Delta(\rho^2) \quad \text{on } (0,\infty) \times \mathbb{R}^d, \qquad \chi > 0. \tag{1.1}$$

It describes local aggregation competing with fourth-order diffusion. The factor ρ in the highest-order flux makes the equation degenerate at vacuum, while the quadratic term is attractive second-order diffusion. A basic qualitative question is whether a compact aggregate retains a bounded edge during its evolution. This does not follow from regularity or moment bounds alone, and a fourth-order equation has no comparison principle of the usual second-order kind.

Carrillo, Esposito, Falcó and Fernández-Jiménez [3] construct global probability-valued weak solutions for a family of such equations. The quadratic attraction in (1.1) lies in their subcritical range in every finite dimension. Their construction gives finite second moments and H^1, H^2 control on finite time intervals. In the Introduction of that work, preservation of compact support is posed as a further question, supported there by numerical observations. We prove this property for (1.1), retaining their weak formulation and data class.

Finite propagation for thin-film equations is closely connected with localized entropy estimates. For example, Novick-Cohen and Shishkov [5] obtain such results for one-dimensional equations with backwards second-order diffusion. Their interval problem and entropy-solution construction differ from the whole-space variational weak solution considered here. The issue is to obtain a sufficiently strong local estimate for that same weak solution, uniformly in the approximations and without an upper restriction on the spatial dimension.

Write $\mathcal{P}_2(\mathbb{R}^d)$ for probability measures with finite second moment. Narrow continuity means continuity against every bounded continuous spatial test. Our main result is the following.

Theorem 1.1. *Let $d \geq 2$ be an integer, $\chi > 0$, and let $\rho_0 \in H^1(\mathbb{R}^d)$ be nonnegative, compactly supported, and have $\int_{\mathbb{R}^d} \rho_0 \, \mathrm{d}x = 1$. There exists one global narrowly continuous curve $\rho : [0,\infty) \to$*

*yui.ui.siki@gmail.com

$\mathcal{P}_2(\mathbb{R}^d)$, *with* $\rho(0)=\rho_0$, *represented by nonnegative probability densities, such that for every finite* $T>0$,

$$\rho \in L^\infty(0,T;H^1(\mathbb{R}^d)) \cap L^2(0,T;H^2(\mathbb{R}^d)).$$

For every $\phi \in C_c^2(\mathbb{R}^d)$ *and every* $0 \le t_1 < t_2 < \infty$,

$$\int \phi[\rho(t_2)-\rho(t_1)]\,\mathrm{d}x = -\int_{t_1}^{t_2}\int [\rho\Delta\rho\Delta\phi + \Delta\rho\nabla\rho\cdot\nabla\phi + \chi\rho^2\Delta\phi]\,\mathrm{d}x\,\mathrm{d}t. \tag{1.2}$$

For each such T *there is* $R_T<\infty$ *for which*

$$\rho(t,x)=0 \quad \text{for almost every } (t,x)\in(0,T)\times(\mathbb{R}^d\setminus B_{R_T}).$$

The same curve has these properties for all finite horizons.

Here B_R is the Euclidean ball centered at zero. The radius may depend on the datum, d,χ and T. The new conclusion concerns the support of a selected global solution; it gives neither uniqueness of the fourth-order weak equation nor a quantitative growth law for R_T. The theorem retains the integrated weak identity and the $L^\infty H^1\cap L^2H^2$ bounds of [3, Definition 2.1] at $m=2$.[1] The support conclusion is established with precisely the solution requirements stated above.

The proof uses a power entropy with exponent slightly above one. Its localized dissipation controls a weighted Hessian and a weighted quartic gradient. Interpolation turns these into a superlinear estimate for the exterior spacetime tail. Because that tail tends to zero far away, an elementary iteration makes it identically zero beyond a finite radius.

At the available weak regularity, direct nonlinear testing does not justify the entropy calculation. Instead we use the flow-interchange principle of Matthes, McCann and Savaré [4] on the original variational approximations. A convex density entropy with bounded derivative admits a spatially weighted metric flow in the sense of Ambrosio, Gigli and Savaré [1]. We establish its energy estimate by renormalization, positive implicit approximation and comparison in a common energy class. The resulting local inequality therefore holds for the same original solution, rather than for a separately chosen entropy solution.

Section 2 introduces these auxiliary entropies and their spatial estimates. Section 3 proves the rough-data auxiliary dissipation. Section 4 obtains the local inequality and proves the theorem. The two interpolation estimates are included in Appendix A. Throughout, $H^j=W^{j,2}(\mathbb{R}^d)$ has its inhomogeneous norm; ∇^2u is the Hessian with Frobenius norm. Integrals without a domain are over $\mathbb{R}^d$, or over the explicitly specified periodic cube in the auxiliary construction. Time suprema are essential suprema. Products with a negative density power are assigned zero at vacuum; the limiting arguments below justify each use.

2 Weighted auxiliary entropies

2.1 The original weak solution and its approximations

Let

$$F(u)=\frac12\int_{\mathbb{R}^d}|\nabla u|^2\,\mathrm{d}x-\chi\int_{\mathbb{R}^d}u^2\,\mathrm{d}x \tag{2.1}$$

for an H^1 probability density, and set $F=+\infty$ otherwise. The minimizing movement construction of Carrillo, Esposito, Falcó and Fernández-Jiménez [3, Definition 2.1, Theorem 2.2 and Section 4]

[1] Definition 2.1(i) of the pinned preprint also writes $p=2^*=\infty$ in dimension two. We do not assert that endpoint. The construction estimates used here are the displayed H^1/H^2 bounds and their consequences for finite spatial exponents.

applies to this energy. In their notation $m = 2 < 2 + 2/d$, for every finite d. Their weak identity has exactly the coefficient χ in (1.2). We recall the estimates of the construction because our localization will be applied to the same minimizers.

Write W_2 for the quadratic Wasserstein distance on $\mathcal{P}_2(\mathbb{R}^d)$,

$$W_2^2(\mu,\nu) = \inf_{\pi\in\Gamma(\mu,\nu)} \int_{\mathbb{R}^d\times\mathbb{R}^d} |x-y|^2 \,\mathrm{d}\pi(x,y),$$

where $\Gamma(\mu,\nu)$ is the set of probability measures with marginals μ,ν. For $\tau > 0$ choose

$$u_k \in \operatorname*{argmin}_{v\in\mathcal{P}_2(\mathbb{R}^d)} \left\{ F(v) + \frac{W_2^2(v,u_{k-1})}{2\tau} \right\}, \qquad u_0 = \rho_0, \tag{2.2}$$

and write $u_\tau(t) = u_k$ on $((k-1)\tau, k\tau]$. For every finite T, the source estimates give

$$\sup_{\tau,t\le T} \left(\|u_\tau(t)\|_{H^1}^2 + \int |x|^2 u_\tau(t)\,\mathrm{d}x \right) \le C_T, \tag{2.3}$$

$$\int_0^T \|u_\tau(t)\|_{H^2}^2 \,\mathrm{d}t \le C_T, \qquad \sum_{k\tau\le T} W_2^2(u_k,u_{k-1}) \le C_T\tau. \tag{2.4}$$

Each u_k has finite entropy and belongs to H^2; the latter follows from the heat-flow estimate in [3, Lemma 4.2]. A subsequence converges to the source weak solution, strongly in spacetime L^2. Interpolating this convergence with (2.4) gives strong convergence in $L^2(0,T;H^1)$. A diagonal choice over integer T gives one global curve. We fix that choice throughout.

Choose the small exponent

$$a = \frac{1}{100(d+1)}, \qquad p = 1+a, \qquad r = 2+3a, \qquad r_1 = 2+4a. \tag{2.5}$$

For $d > 2$ we have $r_1 < 2d/(d-2)$, while for $d = 2$ all these exponents are finite. The usual Sobolev inequality and mass interpolation therefore give uniform L^{r_1} bounds. Interpolating with strong L^2 convergence gives strong spacetime L^r convergence along the same subsequence. Only finite-exponent Sobolev embeddings are used when $d = 2$.

2.2 A bounded derivative and an actual metric flow

The power entropy u^p has the right localization estimates, but its unbounded derivative is inconvenient in a spatially weighted transport flow. We first use a convex approximation with bounded derivative. For $M \ge 1$ and $z \ge 0$ define

$$q_M(z) = \frac{z^a}{1+(z/M)^{2a}}, \qquad f_M'(z) = \frac{M^a}{a}\arctan((z/M)^a), \qquad f_M(0) = 0. \tag{2.6}$$

Thus $f_M''(z) = q_M(z)/z$ for $z > 0$. Put

$$P_M(z) = zf_M'(z) - f_M(z) = \int_0^z q_M(y)\,\mathrm{d}y, \qquad L_M = \frac{\pi M^a}{2a}. \tag{2.7}$$

The elementary bounds needed below are

$$|zq_M'(z)| \le aq_M(z), \qquad 0 \le f_M(z) \le L_M z, \qquad 0 \le f_M'(z) \le L_M, \tag{2.8}$$

$$zq_M(z) \ge (1-a)P_M(z), \qquad zq_M(z) - (1-1/d)P_M(z) \ge (1/d-a)P_M(z), \tag{2.9}$$

$$\frac{f_M'(z)^2}{q_M(z)} \le C_a z^a, \qquad \frac{f_M'(z)^4}{q_M(z)^3} \le C_a(z^a + z^{3a}). \tag{2.10}$$

For (2.9), integrate $q_M(y) \le q_M(z)(z/y)^a$ over $0 < y < z$. For (2.10), on $z \le M$ use $f_M'(z) \le z^a/a$ and $q_M(z) \ge z^a/2$. On $z \ge M$ use $f_M' \le L_M$ and $q_M(z) \ge M^{2a} z^{-a}/2$; the remaining factor M^{-2a} is at most one. Also $q_M \uparrow z^a$ and $f_M \uparrow z^p/(ap)$ as $M \to \infty$.

Let $0 \le \zeta \le 1$ be smooth and constant outside a compact set, and set $\psi = \zeta^4$. For $0 < \epsilon \le 1$ consider

$$G_{M,\epsilon}(u) = \int \psi f_M(u)\,\mathrm{d}x + \epsilon \int u \log u\,\mathrm{d}x. \tag{2.11}$$

Here $0 \log 0 = 0$ and the value is $+\infty$ on singular measures. The weight will eventually vanish on an inner ball and equal one far away. The fourth power makes all derivative quotients used below bounded. For a finite-energy density u, the descending metric slope is

$$|\partial G_{M,\epsilon}|(u) = \limsup_{\substack{v \to u \\ v \ne u}} \frac{[G_{M,\epsilon}(u) - G_{M,\epsilon}(v)]_+}{W_2(u,v)}.$$

Here convergence is in W_2, and $[\cdot]_+$ denotes the positive part.

Proposition 2.1. *For fixed M, ϵ, ψ, the functional (2.11) has an λ_M-gradient-flow semigroup S_s on its finite-energy domain in $\mathcal{P}_2(\mathbb{R}^d)$, where*

$$\lambda_M = -L_M \left(\|\nabla^2 \psi\|_\infty + (1/d - a)^{-1} \| \frac{|\nabla \psi|^2}{\psi} \|_\infty \right). \tag{2.12}$$

The quotient at $\psi = 0$ is assigned zero. The curve $U_s = S_s u$ is W_2-continuous at zero and satisfies the evolution variational inequality

$$\frac{1}{2} \frac{\mathrm{d}}{\mathrm{d}s} W_2^2(U_s, v) + \frac{\lambda_M}{2} W_2^2(U_s, v) \le G_{M,\epsilon}(v) - G_{M,\epsilon}(U_s) \tag{2.13}$$

for almost every $s > 0$ and every finite-energy v. For finite initial energy its tangent velocity v_s satisfies $\partial_s U + \operatorname{div}(Uv) = 0$ and the energy/action identity

$$G_{M,\epsilon}(U_s) + \int_0^s \int |v_t|^2 U_t \,\mathrm{d}x\,\mathrm{d}t = G_{M,\epsilon}(u), \qquad \|v_s\|_{L^2(U_s)} = |\partial G_{M,\epsilon}|(U_s)$$

with the norm equality holding almost everywhere.

Proof. Abbreviate $G = G_{M,\epsilon}$. We check the hypotheses of [1, Theorem 11.2.1]. For the normalized Gaussian γ, nonnegativity of relative entropy gives

$$\int u \log u\,\mathrm{d}x \ge -\frac{1}{2} \int |x|^2 u\,\mathrm{d}x - \frac{d}{2} \log(2\pi). \tag{2.14}$$

Thus G is bounded below on each second-moment ball and a Gaussian is a finite-energy test. These are properness and the literal coercivity condition (11.2.1b) of that theorem.

For completeness the negative entropy tail is uniform on such a ball. Choose $0 < \alpha < 2$ and split $\{0 < u \le 1,\ |x| > R\}$ at $u = \exp(-|x|^\alpha)$. For large R,

$$\int_{|x|>R} (u \log u)^-\,\mathrm{d}x \le R^{\alpha-2} \int |x|^2 u\,\mathrm{d}x + \int_{|x|>R} |x|^\alpha e^{-|x|^\alpha}\,\mathrm{d}x. \tag{2.15}$$

Local convex-integral lower semicontinuity consequently extends to the whole entropy. The nonnegative weighted convex integral in (2.11) is lower semicontinuous as a relaxed integral; the entropy makes its value infinite on singular measures. One may also use local weak L^1 compactness from

the entropy bound and weak lower semicontinuity of the Lipschitz convex functional $u \mapsto \int \psi f_M(u)$; see [2, Corollary 3.9 and Theorem 4.30].

It remains to verify the precise transport convexity required by the source. Let $\mu_1 = m(x)\,\mathrm{d}x$ be a finite-energy base and let T_2, T_3 be its optimal maps to finite-energy endpoints μ_2, μ_3. The generalized geodesic under consideration is

$$T_t = (1-t)T_2 + tT_3, \qquad \mu_t = (T_t)_{\#}\mu_1, \qquad 0 \le t \le 1,$$

where the push-forward notation means $\mu_t(E) = \mu_1(T_t^{-1}(E))$. By [1, Lemma 5.5.3 and Proposition 6.2.12], on a common full base-measure set the maps have positive definite almost-everywhere Jacobians $A_i = DT_i(x)$, are essentially strictly monotone, and obey the density-change formula. Put

$$A_t = (1-t)A_2 + tA_3, \quad D = A_3 - A_2 = D^{\mathsf{T}}, \quad J_t = \det A_t > 0.$$

The interpolated map remains essentially injective. At $y = T_t(x)$ its density, transported velocity and deformation matrix are respectively

$$z_t(y) = m(x)/J_t, \qquad v_t(y) = T_3(x) - T_2(x), \qquad B_t(y) = DA_t^{-1}.$$

Pulling the entropy back to the base gives the scalar integrand

$$K_x(t) = J_t\psi(T_t(x))f_M(m(x)/J_t) + \epsilon m(x)\log(m(x)/J_t).$$

For regular transport data, the second variation of the weighted energy is

$$\int [\psi\{(zq_M - P_M)(\operatorname{tr} B)^2 + P_M \operatorname{tr}(B^2)\} - 2P_M \operatorname{tr} B\, \nabla\psi\cdot v + f_M\nabla^2\psi[v,v]]\,\mathrm{d}y, \tag{2.16}$$

where $z = z_t(y)$, $v = v_t(y)$, $B = B_t(y)$, and the scalar functions are evaluated at z. The matrix B is similar to the symmetric matrix $A_t^{-1/2}DA_t^{-1/2}$, so $\operatorname{tr}(B^2) \ge (\operatorname{tr} B)^2/d$. By (2.9) and completion of a square, (2.16) is at least $\lambda_M \int z|v|^2\,\mathrm{d}y$. The entropy has nonnegative second variation.

The same algebra applies pointwise to K_x on the common full set, before any differentiation is passed through a spatial integral. It yields $K_x''(t) \ge \lambda_M m(x)|T_3(x) - T_2(x)|^2$ also for the nonsmooth transport maps. Integrating first in t and then in x yields

$$G(\mu_t) \le (1-t)G(\mu_2) + tG(\mu_3) - \frac{\lambda_M}{2}t(1-t)\int |T_3 - T_2|^2\,\mathrm{d}\mu_1. \tag{2.17}$$

The endpoint energies and induced cost are finite; the negative entropy part is integrable by (2.15). This justifies the integration also before positive intermediate entropy integrability is known.

The cost in (2.17) is the induced three-plan cost. For negative λ_M it is retained in that form. This is exactly [1, Definition 9.2.4 and (11.2.1a)]; the distinction from endpoint cost is explained in its Remark 9.2.6. The generation theorem itself yields the endpoint-distance inequality (2.13) and the energy/action and slope identities (11.2.4), (11.2.9). □

2.3 A weighted spatial coercivity estimate

We next compute the decay of F for the auxiliary equation. Its rigorous application to the rough curve $S_s u$ occupies Section 3. For a positive smooth periodic density write $q = q_M(u)$, $f' = f_M'(u)$ and

$$\mathcal{C}_\psi(u) = \int \psi q|\nabla^2 u|^2, \qquad \mathcal{D}_\psi(u) = \int \psi\frac{q}{u^2}|\nabla u|^4. \tag{2.18}$$

The formulas below also define these nonnegative quantities on rough whole-space densities, with vacuum values zero. Set

$$E_M(u) = C_d \int u^2 \left[\left(q + \frac{f'^4}{q^3} \right) \frac{|\nabla\psi|^4}{\psi^3} + \frac{f'^2}{q} \frac{|\Delta\psi|^2}{\psi} \right], \tag{2.19}$$

$$B_M(u) = E_M(u) + C_d\chi^2 \int \psi q_M(u) u^2 + C_d\chi \int u^{2+a} |\Delta\psi|. \tag{2.20}$$

Here and below a sufficiently large C_d can be fixed once; it is independent of M, ϵ and the periodic box. These errors have growth through $u^{2+a} + u^{2+3a}$, by (2.10).

Lemma 2.2. *Let ψ be constant near the boundary of a periodic cube and let $u > 0$ be smooth and periodic. For*

$$J(u) = (\epsilon + \psi q_M(u))\nabla u + u f'_M(u) \nabla\psi, \tag{2.21}$$

there is $c_d > 0$, independent of M, ϵ and the cube, such that

$$-\int (-\Delta u - 2\chi u) \operatorname{div} J(u) \geq c_d(\mathcal{C}_\psi(u) + \mathcal{D}_\psi(u)) + \frac{\epsilon}{2}\|\Delta u\|_2^2 - B_M(u) - 2\chi\epsilon\|\nabla u\|_2^2. \tag{2.22}$$

Proof. Put $g = \nabla u$ and $H = \nabla^2 u$. Integrating $\operatorname{div}[\psi(q/u)|g|^2 g]$ and using $|uq'| \leq aq$ gives

$$(1-a)\mathcal{D}_\psi \leq (2+\sqrt{d})\sqrt{\mathcal{C}_\psi \mathcal{D}_\psi} + \mathcal{D}_\psi^{3/4} \left(\int q u^2 \frac{|\nabla\psi|^4}{\psi^3} \right)^{1/4}.$$

Consequently

$$\mathcal{D}_\psi \leq \frac{4(2+\sqrt{d})^2}{(1-a)^2} \mathcal{C}_\psi + C_d \int q u^2 \frac{|\nabla\psi|^4}{\psi^3}. \tag{2.23}$$

A second integration by parts gives

$$\int \psi \{ q(\Delta u)^2 + q'|g|^2 \Delta u \} = \mathcal{C}_\psi + \int \psi q' H[g,g] + \int q \nabla\psi \cdot Hg - \int q \Delta u \nabla\psi \cdot g. \tag{2.24}$$

Expanding $\int \Delta u \operatorname{div} J(u)$ and using (2.24), its non-heat terms are

$$\mathcal{C}_\psi + \int \psi q' H[g,g] + \int q \nabla\psi \cdot Hg + \int (f' + q) \Delta u \nabla\psi \cdot g + \int u f' \Delta\psi \Delta u. \tag{2.25}$$

The first error is at least $-a\sqrt{\mathcal{C}_\psi \mathcal{D}_\psi}$. Our choice of a satisfies $2a(2+\sqrt{d})/(1-a) < 1/4$. Combining (2.23) with Young's inequality retains a fixed positive multiple of $\mathcal{C}_\psi + \mathcal{D}_\psi$. The next two terms in (2.25) are first absorbed into $\mathcal{C}_\psi$ and then into $\mathcal{D}_\psi$. Their remainders are bounded by $C_d u^2 (q + f'^4/q^3)|\nabla\psi|^4/\psi^3$ after integration. The final term leaves $C_d u^2 f'^2 |\Delta\psi|^2/(q\psi)$. These are exactly the terms in (2.19).

The attraction contributes

$$-2\chi \int \psi q |g|^2 - 2\chi \int u f' \nabla\psi \cdot g.$$

The first term is absorbed into a fraction of $\mathcal{D}_\psi$ with remainder $C_d\chi^2 \int \psi q u^2$. For the second use $H_M(z) = \int_0^z y f'_M(y)\,dy \leq z^{2+a}/[a(2+a)]$. It equals $2\chi \int H_M(u)\Delta\psi$ and is bounded below by the last error in (2.20). The heat contribution is $\epsilon\|\Delta u\|_2^2 - 2\chi\epsilon\|\nabla u\|_2^2$. Decreasing c_d once and retaining half the positive heat term proves (2.22). □

3 Dissipation along the auxiliary flow

Fix M, ϵ, ψ for this section and abbreviate $G = G_{M,\epsilon}$. The preceding spatial calculation must be established along the actual flow generated by Proposition 2.1, starting from a rough density. We prove the following statement.

Proposition 3.1. *Let $\bar{u} \in H^1(\mathbb{R}^d)$ be a nonnegative probability density in $\mathcal{P}_2(\mathbb{R}^d)$, and let $U_s = S_s\bar{u}$. For every finite S, this curve belongs to $L^\infty(0,S;H^1) \cap L^2(0,S;H^2)$ and satisfies, for every $s \geq 0$,*

$$F(U_s) + \int_0^s \left[c_d(\mathcal{C}_\psi(U_t) + \mathcal{D}_\psi(U_t)) + \frac{\epsilon}{2}\|\Delta U_t\|_2^2 \right] \mathrm{d}t \leq F(\bar{u}) + \int_0^s [B_M(U_t) + 2\chi\epsilon\|\nabla U_t\|_2^2]\,\mathrm{d}t. \quad (3.1)$$

The constants and errors are those of Lemma 2.2. Moreover $U_s \to \bar{u}$ strongly in H^1 and L^r as $s \downarrow 0$, and

$$\liminf_{s\downarrow 0} \frac{F(\bar{u}) - F(S_s\bar{u})}{s} \geq c_d(\mathcal{C}_\psi(\bar{u}) + \mathcal{D}_\psi(\bar{u})) + \frac{\epsilon}{2}\|\Delta\bar{u}\|_2^2 - B_M(\bar{u}) - 2\chi\epsilon\|\nabla\bar{u}\|_2^2. \quad (3.2)$$

The positive terms on the right may initially have value $+\infty$.

We first obtain a common energy class for the actual metric flow. We then construct positive periodic approximations for which the spatial estimate is legitimate. A direct comparison in H^{-1} identifies their limit with the metric flow and permits the energy inequality to pass to it.

3.1 The flux and a renormalized identity

Set

$$\mathcal{A}(x,z) = \epsilon z + \psi(x)P_M(z), \qquad \kappa(x,z) = \epsilon + \psi(x)q_M(z). \quad (3.3)$$

Thus $\epsilon \leq \partial_z\mathcal{A} = \kappa \leq \Lambda$, where $\Lambda = \epsilon + \|q_M\|_\infty$.

Lemma 3.2. *The curve $U = S_s\bar{u}$ satisfies*

$$\partial_s U = \operatorname{div} J, \qquad J = \kappa(x,U)\nabla U + U f_M'(U)\nabla\psi = \nabla\mathcal{A}(x,U) + f_M(U)\nabla\psi. \quad (3.4)$$

On every finite interval, both $J/\sqrt{U}$ and $\nabla U/\sqrt{U}$ belong to spacetime L^2, with value zero at vacuum.

Proof. At almost every positive time the metric-flow velocity v is tangent, $\eta = -v$ is a subgradient, and $\|\eta\|_{L^2(U)} = |\partial G|(U) =: \sigma$. These are the conclusions of [1, Definition 11.1.1 and Theorem 11.2.1]. At an absolutely continuous measure, the reduced subgradient agrees with the ordinary one by its Remark 10.3.3. For clarity, at an absolutely continuous finite-slope density the ordinary subgradient condition is [1, Definition 10.1.1, (10.1.3)]

$$G(\nu) - G(U) \geq \int \eta(x)\cdot(X(x)-x)U(x)\,\mathrm{d}x + o(W_2(U,\nu)) \quad \text{as } \nu \to U,$$

where X is the optimal transport map from U to the finite-energy measure ν, and convergence is in W_2. Only optimal perturbations will be used in this inequality. The tangent space is the $L^2(U)$ closure of compact smooth gradients; the cylindrical formulation in [1, Definition 8.4.1] is equivalent in $\mathbb{R}^d$ by multiplying a bounded cylindrical function by cutoffs tending to one. The gradient error tends to zero in $L^2(U)$.

For $\xi \in C_c^\infty(\mathbb{R}^d;\mathbb{R}^d)$, direct change of variables gives

$$\mathcal{L}(\xi) := \frac{\mathrm{d}}{\mathrm{d}h} G((\mathrm{Id}+h\xi)_\# U)\Big|_{h=0} = \int f_M(U)\nabla\psi\cdot\xi - \int [\psi P_M(U) + \epsilon U]\,\mathrm{div}\,\xi. \tag{3.5}$$

Both signs of small h give diffeomorphisms. The derivative is dominated by an integrable multiple of U, using $f_M \le L_M U$, $P_M \le L_M U$ and the bounded logarithmic Jacobian for the entropy. The metric descending-slope definition and $W_2(U,(\mathrm{Id}+h\xi)_\# U) \le |h|\|\xi\|_{L^2(U)}$ give

$$|\mathcal{L}(\xi)| \le \sigma\|\xi\|_{L^2(U)}.$$

Choose the sign decreasing the differentiable energy to obtain this bound. If the norm is zero, the pushforward is the identity U-almost everywhere. By density and Riesz representation, $\mathcal{L}(\xi) = \int z\cdot\xi\,U$ for some $z \in L^2(U)$ with $\|z\| \le \sigma$.

For $\xi = \nabla\phi$, $\phi \in C_c^\infty$, the map $\mathrm{Id} + h\nabla\phi$ is optimal for both sufficiently small signs: it is the gradient of the convex function $|x|^2/2 + h\phi$. Fenchel's inequality proves optimality, since equality holds on its graph and both the potential and its conjugate differ from a quadratic by a bounded function. Their marginal integrals are finite in $\mathcal{P}_2(\mathbb{R}^d)$. The ordinary subgradient inequality can therefore be applied for those gradient perturbations and yields $\mathcal{L}(\nabla\phi) = \int \eta\cdot\nabla\phi\,U$. The tangent projection of z is consequently η. Since $\|\eta\| = \sigma \ge \|z\|$, orthogonal norm decomposition forces $z = \eta$. Thus, for all vector tests, (3.5) says

$$U\eta = \nabla[\epsilon U + \psi P_M(U)] + f_M(U)\nabla\psi.$$

The velocity is $v = -\eta$, so its continuity equation has exactly the sign in (3.4). The energy/action identity gives $J/\sqrt{U} \in L^2$. Unit mass then gives $J \in L^1$ in spacetime.

The increasing inverse $\mathcal{T}(x,V)$ of $\mathcal{A}(x,z)$ has $|\mathcal{T}_V| \le \epsilon^{-1}$ and $|\mathcal{T}_x| \le C|V|$. Since $\mathcal{A}(x,U)$ and its distributional gradient are L^1, the spatial chain rule, first on truncated V and then by these integrable bounds, gives $U \in W^{1,1}$ in space and the first flux formula in (3.4). At vacuum both J and ∇U vanish almost everywhere. Finally

$$\int_0^S\int \frac{|\nabla U|^2}{U} \le \frac{2}{\epsilon^2}\left[\int_0^S\int \frac{|J|^2}{U} + SL_M^2\|\nabla\psi\|_\infty^2\right] < \infty. \tag{3.6}$$

□

Lemma 3.3. *If $\ell \ge 2$ and $\bar{u} \in L^\ell$, then for every $0 \le s \le t \le S$,*

$$\|U_t\|_\ell^\ell + \ell(\ell-1)\int_s^t\int U^{\ell-2}\kappa(x,U)|\nabla U|^2 = \|U_s\|_\ell^\ell + \int_s^t\int H_\ell(U)\Delta\psi, \tag{3.7}$$

where $H_\ell(z) = \ell(\ell-1)\int_0^z y^{\ell-1}f_M'(y)\,\mathrm{d}y$. In particular

$$\|U_t\|_\ell^\ell \le e^{(\ell-1)L_M\|\Delta\psi\|_\infty t}\|\bar{u}\|_\ell^\ell. \tag{3.8}$$

Thus $U \in L^\infty L^2 \cap L^2H^1$, its flux is L^2, and $\partial_s U \in L^2H^{-1}$ on finite intervals.

Proof. The time chain rule is proved before removing a density truncation. Define

$$\beta_n(z) = \begin{cases} z^\ell, & 0 \le z \le n, \\ \ell n^{\ell-1}z - (\ell-1)n^\ell, & z > n. \end{cases}$$

Its first derivative and $z\beta_n''(z)$ are bounded for fixed n; the latter has only one jump, at n. Put $H_n(z) = \int_0^z y f_M'(y)\beta_n''(y)\,\mathrm{d}y$. We claim the exact identity

$$\int \beta_n(U_t) + \int_s^t \int \beta_n''(U)\kappa(x,U)|\nabla U|^2 = \int \beta_n(U_s) + \int_s^t \int H_n(U)\Delta\psi. \tag{3.9}$$

Here are the convolution and trace details. Let η_δ be a nonnegative spatial mollifier and $U_\delta = \eta_\delta * U$. For any V vanishing at vacuum with $|V|^2/U \in L^1$, weighted Cauchy–Schwarz gives

$$\frac{|\eta_\delta * V|^2}{U_\delta} \le \eta_\delta * \left(\frac{|V|^2}{U}\right). \tag{3.10}$$

Choose a common subsequence of approximate identities converging almost everywhere for U, V and $|V|^2/U$. On $\{U > 0\}$ the quotient vector converges to $V/\sqrt{U}$. On vacuum its square is bounded by a quantity tending to zero in (3.10). Fatou and the integrated upper bound show convergence of its L^2 norm. Weak compactness identifies its weak limit, hence the convergence is strong. Applying this to J and ∇U, allowed by Lemma 3.2, gives

$$\frac{J_\delta}{\sqrt{U_\delta}} \longrightarrow \frac{J}{\sqrt{U}}, \qquad \frac{\nabla U_\delta}{\sqrt{U_\delta}} \longrightarrow \frac{\nabla U}{\sqrt{U}} \quad \text{strongly in spacetime } L^2. \tag{3.11}$$

For fixed δ, $\partial_s U_\delta = (\nabla\eta_\delta) * J \in L^1$ in spacetime. Thus U_δ has an absolutely continuous L^1 representative. It agrees at every time with $\eta_\delta * U_s$: both are distributionally continuous in time, the latter by narrow continuity. Its initial value is exactly $\eta_\delta * \bar{u}$. One may now convolve in time on an interior interval, apply the scalar chain rule for β_n, and pass back. The derivative difference is L^1, and the remaining factor converges against that fixed derivative because β_n' is bounded and continuous. The Lipschitz bound of β_n passes endpoints. Absolute continuity extends the identity to $s = 0$ and $t = S$.

Multiply the smoothed equation by $\chi_R\beta_n'(U_\delta)$, where χ_R is a compact cutoff with $|\nabla\chi_R| \le C/R$, and integrate in space and time. The boundary term is at most $CR^{-1}\|\beta_n'\|_\infty \int_0^S \int |J|$ and vanishes. The other integrand is

$$[U_\delta\beta_n''(U_\delta)]\frac{\nabla U_\delta}{\sqrt{U_\delta}} \cdot \frac{J_\delta}{\sqrt{U_\delta}}.$$

The product of the weighted vectors converges strongly in L^1 by (3.11). Away from $U = n$ its bounded multiplier converges pointwise. On $U = n$, $\nabla U = 0$ almost everywhere by Sobolev locality, so the limiting product is zero and the jump leaves no error. This proves strong L^1 convergence of the integrand. At each endpoint, spatial convolution converges in L^1 for its actual density. Consequently

$$\int \beta_n(U_t) - \int \beta_n(U_s) = -\int_s^t \int \beta_n''(U)\nabla U \cdot J$$

for every endpoint pair, on the same convolution subsequence. Substitution of (3.4) and $\nabla H_n(U) = U f_M'(U)\beta_n''(U)\nabla U$ gives (3.9) by integration against compactly supported $\nabla\psi$.

The estimate uniform in n is

$$0 \le H_n(z) \le L_M[z\beta_n'(z) - \beta_n(z)] \le L_M(\ell - 1)\beta_n(z).$$

Below n the pressure is $(\ell-1)z^\ell$; above n it is $(\ell-1)n^\ell$, while $\beta_n(z) \ge n^\ell$. Gronwall first gives $\int \beta_n(U_t) \le e^{(\ell-1)L_M\|\Delta\psi\|_\infty t} \int \bar{u}^\ell$. The functions β_n increase to z^ℓ, so monotone convergence proves (3.8) and spacetime L^ℓ integrability. Now $H_n(U) \le L_M(\ell-1)U^\ell$ is an integrable majorant for the drift. The positive diffusion integrand increases at positive densities. Dominated and monotone convergence in (3.9) prove (3.7). Taking $\ell = 2$ proves the stated energy class; bounded coefficients in the flux then give its L^2 bound and the H^{-1} time derivative. Taking $\ell = r_1$ is justified by (2.5) and the initial H^1 norm. □

3.2 Comparison in the energy class

We will construct a second solution of (3.4) with an energy inequality. The following elementary comparison identifies it.

Lemma 3.4. *Two nonnegative narrowly continuous probability-density solutions of*

$$\partial_s U = \Delta \mathcal{A}(x, U) + \operatorname{div}(f_M(U)\nabla\psi)$$

with the same initial measure and in $L^\infty L^2 \cap L^2 H^1$ on finite intervals coincide.

Proof. For two solutions U, W put $w = U - W$, $C = \mathcal{A}(x, U) - \mathcal{A}(x, W)$, and $D = (f_M(U) - f_M(W))\nabla\psi$. The coefficient bounds give $C \in L^2 H^1$, $D \in L^2$, and $w_s = \Delta C + \operatorname{div} D \in L^2 H^{-1}$. Thus w is an H^1 time curve in the Hilbert space H^{-1}. Its continuous representative has zero initial trace: narrow continuity identifies that trace in distributions.

Define the inhomogeneous resolvent $R = (I - \Delta)^{-1}$ variationally by

$$Z = Rw \in H^1, \qquad \int Z\phi + \nabla Z \cdot \nabla\phi = \langle w, \phi\rangle \quad (\phi \in H^1).$$

This is Riesz representation for the H^1 inner product. For $E = \|w\|_{H^{-1}}^2 = \|Z\|_2^2 + \|\nabla Z\|_2^2$, the Hilbert time chain rule and the resolvent equation tested with C give

$$\frac{E'}{2} = -\int \nabla C \cdot \nabla Z - \int D \cdot \nabla Z = \int CZ - \int Cw - \int D \cdot \nabla Z.$$

Monotonicity and Lipschitzness yield $\int Cw \geq \epsilon\|w\|_2^2$, $\|C\|_2 \leq \Lambda\|w\|_2$ and $\|D\|_2 \leq L_M\|\nabla\psi\|_\infty\|w\|_2$. Therefore

$$\frac{E'}{2} \leq -\frac{\epsilon}{2}\|w\|_2^2 + \frac{(\Lambda + L_M\|\nabla\psi\|_\infty)^2}{2\epsilon} E.$$

Since $E(0) = 0$, Gronwall gives $w = 0$. Narrow continuity identifies the representatives at every time. □

3.3 Positive implicit approximations

Let $Q_L = (-L, L)^d$ be periodic, with $L \geq 1$ large enough that ψ is constant near its boundary. The derivatives of its periodic restriction have the original bounds. Extend P_M, q_M, f_M by zero for negative arguments, so $\mathcal{A}(x, z) = \epsilon z$ there. The increasing inverse $\mathcal{T}$ has slopes between Λ^{-1} and ϵ^{-1}. Also $B(z) := z f_M'(z)$ is globally Lipschitz with constant at most $L_M + \|q_M\|_\infty$.

Given a nonnegative periodic previous density $w \in L^2(Q_L)$, an implicit step is

$$u - h\Delta\mathcal{A}(x, u) - h \operatorname{div}(f_M(u)\nabla\psi) = w. \tag{3.12}$$

Here h is an auxiliary construction step. It is separate from the original minimizing movement step τ in (2.2).

Lemma 3.5. *Let $w \in L^2(Q_L)$ be nonnegative and periodic. For all sufficiently small $h > 0$, depending only on the fixed M, ϵ, ψ, equation (3.12) has a unique periodic H^1 solution. It is nonnegative and preserves mass. If $0 < k_{\text{old}} \leq w \leq K_{\text{old}} < \infty$, its solution satisfies*

$$\frac{k_{\text{old}}}{1 + hC_0} \leq u \leq \frac{K_{\text{old}}}{1 - hC_0}, \qquad C_0 = L_M\|\Delta\psi\|_\infty, \tag{3.13}$$

and belongs to $H^2(Q_L)$. The restriction on h is independent of $L, k_{\text{old}}, K_{\text{old}}$.

Proof. Use the variable $V = \mathcal{A}(x,u)$ and the Hilbert norm $\|V\|_h^2 = \|V\|_2^2 + h\|\nabla V\|_2^2$ on periodic H^1. The transformed operator is

$$\langle K_h(V), \phi\rangle = \int \mathcal{T}(x,V)\phi + h\nabla V\cdot\nabla\phi + hf_M(\mathcal{T}(x,V))\nabla\psi\cdot\nabla\phi.$$

For $D = V - W$ its monotonicity estimate is

$$\begin{aligned}\langle K_h(V) - K_h(W), D\rangle &\geq \left[\frac{1}{\Lambda} - \frac{hL_M^2\|\nabla\psi\|_\infty^2}{2\epsilon^2}\right]\|D\|_2^2 + \frac{h}{2}\|\nabla D\|_2^2\\ &\geq c\|D\|_h^2, \qquad c = \min\{1/(2\Lambda), 1/2\} > 0\end{aligned}$$

for small h. Its dual Lipschitz constant in this norm is at most $K = \epsilon^{-1} + 1 + \sqrt{h}L_M\|\nabla\psi\|_\infty/\epsilon$. Let $R_h = I - h\Delta$ be the Riesz map for this Hilbert norm. The map $V \mapsto V - \delta R_h^{-1}(K_h(V) - w)$ is a contraction for $0 < \delta < 2c/K^2$, as follows by expanding its squared norm. The contraction theorem gives the solution of $K_h(V) = w$. The inverse chain gives $u \in H^1$, because $\mathcal{T}_V$ is bounded and $|\mathcal{T}_x(x,V)| \leq C|V|$.

Testing with $\min(V, 0)$ eliminates the drift on the negative set, where $\mathcal{T} = V/\epsilon$. The resulting nonnegative sum is at most zero, so $V, u \geq 0$. Testing with one gives mass conservation. The step can also be written as

$$u - h\,\mathrm{div}[\kappa(x,u)\nabla u + B(u)\nabla\psi] = w.$$

The upper constant in (3.13) is a supersolution and the lower constant is a subsolution. Subtract the supersolution and test with $z = (u - K)_+$. With $K_B = L_M + \|q_M\|_\infty$,

$$\|z\|_2^2 + h\epsilon\|\nabla z\|_2^2 \leq hK_B\|\nabla\psi\|_\infty\|z\|_2\|\nabla z\|_2.$$

Small h and Young's inequality force $z = 0$. The lower barrier follows by the negative-part test. Thus on $jh \leq S$ the successive densities stay between $k_{\text{initial}}e^{-C_0S}$ and $K_{\text{initial}}e^{2C_0S}$, provided $hC_0 \leq 1/2$.

The transformed equation gives

$$-\Delta V = (w - u)/h + \mathrm{div}(f_M(u)\nabla\psi) \in L^2.$$

Periodic Fourier expansion gives $V \in H^2$, and $0 \leq V \leq \Lambda u$ makes it bounded. For any bounded periodic H^2 function V,

$$\|\nabla V\|_4^2 \leq (2 + \sqrt{d})\|V\|_\infty\|\nabla^2 V\|_2. \tag{3.14}$$

For smooth V this follows by integrating $\mathrm{div}(V|\nabla V|^2\nabla V)$ and applying Cauchy–Schwarz. Periodic convolution, preserving the L^∞ bound, proves the general case. On the positive bounded range of u the inverse $\mathcal{T}$ has bounded derivatives through order two. Its second derivative is now legitimate since $\nabla^2 V$ and $|\nabla V|^2$ are L^2. Hence $u \in H^2$, and (3.14) also gives $\nabla u \in L^4$. Positive periodic convolution then extends every identity in Lemma 2.2 to these steps. In particular, the gradients converge in L^4: apply (3.14) to the convolution difference, whose L^∞ norm is bounded and whose Hessian tends to zero in L^2. □

The estimates needed to remove the boxes and positive floors do not use their maximum and minimum. For each fixed $\ell \geq 2$, convexity and testing the step with $\ell u^{\ell-1}$ give

$$\begin{aligned}\|u\|_\ell^\ell - \|w\|_\ell^\ell + h\ell(\ell-1)\int u^{\ell-2}\kappa(x,u)|\nabla u|^2 &\leq h\int H_\ell(u)\Delta\psi\\ &\leq h(\ell-1)L_M\|\Delta\psi\|_\infty\|u\|_\ell^\ell.\end{aligned} \tag{3.15}$$

For small h, iteration gives the uniform bound $\sup_{jh\le S}\|u_j\|_\ell^\ell \le e^{2(\ell-1)L_M\|\Delta\psi\|_\infty S}\|u_0\|_\ell^\ell$. In particular, $\ell=2$ gives

$$\sum_{jh\le S} h\|\nabla u_j\|_2^2 \le C_S, \qquad \sum_{jh\le S} h\|J(u_j)\|_2^2 \le C_S. \tag{3.16}$$

For moment tightness use the genuinely periodic test

$$Z_L(x)=\sum_{i=1}^{d}(2L/\pi)^2\sin^2(\pi x_i/(2L)).$$

On Q_L it satisfies $(4/\pi^2)|x|^2 \le Z_L(x) \le |x|^2$, $|\Delta Z_L| \le 2d$ and $|\nabla Z_L| \le 2|x|$. Equation (3.12), $\mathcal{A}(x,u) \le \Lambda u$ and mass one imply

$$\int Z_L(u_j-u_{j-1}) \le h\left[2d\Lambda + 2L_M \sup_{\operatorname{supp}\nabla\psi}|x||\nabla\psi(x)|\right]. \tag{3.17}$$

The physical second moments of the densities restricted to Q_L are therefore uniformly bounded. Convexity of the entropy and the test $1+\log u_j$ give

$$H(u_j)-H(u_{j-1})+h\epsilon\int\frac{|\nabla u_j|^2}{u_j} \le h\int f_M(u_j)\Delta\psi \le hL_M\|\Delta\psi\|_\infty. \tag{3.18}$$

Their zero extensions, considered only as densities and not Sobolev extensions, obey (2.14). Thus (3.17) and bounded initial entropy give uniform Fisher information summed in time, and also

$$\sum_{jh\le S} h\int\frac{|J(u_j)|^2}{u_j} \le C_S. \tag{3.19}$$

We retain the time-discretization defect in the nonconvex energy F. Let $z=u-w=h\operatorname{div}J(u)$, which has mean zero. Periodic Fourier duality gives

$$\|z\|_2^2 \le \|z\|_{\dot H^{-1}}\|\nabla z\|_2, \qquad \|z\|_{\dot H^{-1}} \le h\|J(u)\|_2.$$

Here the homogeneous norm uses frequencies $\pi n/L$, with the zero mode omitted. The exact quadratic identity yields

$$\begin{aligned} F(u)-F(w) &= \langle -\Delta u-2\chi u, z\rangle - \frac{1}{2}\|\nabla z\|_2^2+\chi\|z\|_2^2 \\ &\le h\langle -\Delta u-2\chi u, \operatorname{div}J(u)\rangle + \chi^2h^2\|J(u)\|_2^2. \end{aligned} \tag{3.20}$$

The pairing is an L^2 pairing: the step equation gives $\operatorname{div}J=(u-w)/h\in L^2$ and $u\in H^2$. Lemma 2.2 now gives

$$F(u)+h\left[c_d(\mathcal{C}_\psi(u)+\mathcal{D}_\psi(u))+\frac{\epsilon}{2}\|\Delta u\|_2^2\right] \le F(w)+h[B_M(u)+2\chi\epsilon\|\nabla u\|_2^2]+\chi^2h^2\|J(u)\|_2^2. \tag{3.21}$$

The sum of the final terms is $O(h)$ by (3.16). The density errors are bounded using (3.15) with $\ell=r_1$, and the gradient error by (3.16). The periodic mass-one coercivity estimate

$$F(u) \ge \tfrac{1}{4}\|\nabla u\|_2^2 - C_{d,\chi}, \qquad L\ge 1, \tag{3.22}$$

proved in Lemma A.2, is uniform in the box. Summing (3.21) gives uniform $\sup H^1$, $\sum h\|u_j\|_{H^2}^2$ and weighted positive dissipation bounds. All constants in these construction bounds may depend on the fixed M,ϵ,ψ,S; none depends on the floor or box size.

3.4 Whole-space convergence and the initial dissipation

Choose $L_j \to \infty$ and smooth compactly supported probability densities ρ_j, supported well inside Q_{L_j}, converging to $\bar{u}$ in H^1, L^{r_1} and second moment. Smooth expanding cutoffs, normalization, and a sufficiently fine centered convolution give these approximations. Set $\alpha_j = L_j^{-4}$ and

$$u_{0,j} = (1-\alpha_j)\rho_j + \frac{\alpha_j}{(2L_j)^d}.$$

The floors have second moment $O(\alpha_j L_j^2) \to 0$, zero gradient and vanishing L^2 norm. Thus $F(u_{0,j}) \to F(\bar{u})$; initial moments, L^{r_1} norms and entropy bounds are uniform. Choose $h_j \to 0$ small enough for the preceding restrictions and construct steps through $S+1$. Write u_j for their piecewise constant density, J_j for its flux, and v_j for the piecewise affine interpolation. As fields or measures only, extend them by zero outside the physical cube. The estimates above are

$$\begin{aligned} &\sup_{j,t}\left[\|u_j(t)\|_{H^1(Q_{L_j})} + \|u_j(t)\|_{r_1} + \int |x|^2 u_j(t)\,\mathrm{d}x\right] \le C, \\ &\int_0^{S+1}\int_{Q_{L_j}} \left[|\nabla^2 u_j|^2 + |J_j|^2 + \frac{|J_j|^2}{u_j}\right]\mathrm{d}x\,\mathrm{d}t \le C. \end{aligned} \tag{3.23}$$

Both interpolations have mass one and the same uniform moment and spatial norm bounds, while $\partial_t v_j = \operatorname{div} J_j$ on each step. For a compact smooth test ϕ, eventually supported inside the cube,

$$\left|\int \phi[v_j(t) - v_j(s)]\right| \le \|\nabla\phi\|_\infty \sqrt{|t-s|}\left(\int_s^t \int \frac{|J_j|^2}{u_j}\right)^{1/2}.$$

Tightness and a diagonal Arzelà argument for a countable determining family of tests give one narrowly continuous probability curve W_t with initial density $\bar{u}$ and bounded second moment.

Here the local strong convergence can be seen directly. Fix a smooth compact cutoff η and place ηv_j on one fixed periodic box. It is bounded in L^2H^1, and $\partial_t(\eta v_j) = \operatorname{div}(\eta J_j) - \nabla\eta \cdot J_j$ is bounded in L^2H^{-1}. The spacetime L^2 norm squared of Fourier modes above frequency K is at most CK^{-2}. Each retained coefficient is bounded in time H^1 and hence has a uniform one-half Hölder modulus. Finite-dimensional compactness followed by $K \to \infty$ proves local strong L^2 convergence. On each step $\|v_j - u_j\|_{H^{-1}_{\mathrm{loc}}} \le C h_j \|J_j\|_2$. The Fourier interpolation inequality between H^{-1} and H^1 shows that u_j has the same local strong L^2 limit. The local L^2H^2 bound then gives strong local gradients. A subsequence converges almost everywhere in density and gradient. At every fixed time the uniform H^1, L^{r_1} bounds and the prescribed narrow limit give a density W_t with those same bounds.

To pass the energy errors, local convergence is insufficient. Put $\vartheta = (r_1-2)/(r_1-1) > 0$. Mass interpolation and the moment bound give

$$\int_{|x|>R} u_j^2 \le \left(\int_{|x|>R} u_j\right)^{\vartheta}\left(\int u_j^{r_1}\right)^{1/(r_1-1)} \le CR^{-2\vartheta}. \tag{3.24}$$

The same holds for W, so the densities converge strongly in global spacetime L^2 and, by interpolation, in L^r. For gradients choose $\theta_R = 0$ on $B_{R/2}$ and $\theta_R = 1$ outside B_R, with $|\nabla\theta_R| \le C/R$. When $L_j > 2R$ it is constant near the periodic boundary. Periodic integration by parts gives

$$\int \theta_R^2 |\nabla u_j|^2 \le 2\|\theta_R u_j\|_2 \|\theta_R \Delta u_j\|_2 + CR^{-2}\int_{|x|>R/2} u_j^2. \tag{3.25}$$

After time integration, (3.23) and (3.24) make this tail tend to zero uniformly. The Hessian fields, extended as L^2 fields, have a global weak limit identified with $\nabla^2 W$ by interior compact tests. Hence $W \in L^2H^2$ and the analogous gradient-tail bound holds for it. Combining local gradient convergence with these tails gives global strong gradient convergence. This reasoning does not differentiate a zero extension.

Bounded continuity of q_M and Lipschitzness of $B(z) = zf'_M(z)$ now give

$$J_j \longrightarrow \kappa(x, W)\nabla W + Wf'_M(W)\nabla\psi \quad \text{strongly in global spacetime } L^2.$$

Pass the compact-test equation to obtain (3.4) for W, with its initial measure. Both W and the actual U belong to the class of Lemma 3.4; thus $W = U$ at every time.

Sum (3.21) and pass to this limit. At almost every endpoint t, the densities converge in global L^2 and the gradients are weakly compact, so $F(U_t) \le \liminf F(u_j(t))$. For the positive spacetime terms, $\sqrt{q_M(u_j)}$ is bounded and converges almost everywhere; multiplying an L^2 test therefore converges strongly, which gives weighted Hessian lower semicontinuity. Fatou passes the quartic term on $\{U > 0\}$, and its target value at vacuum is zero. The heat term is weakly lower semicontinuous.

Each density integrand in B_M is continuous at zero, has growth bounded by $C(z^{2+a} + z^{2+3a})$, and converges in L^1 by the global strong L^2, L^r convergence. On bounded density ranges its derivative is bounded by $C_K z$; the remaining tails are uniformly integrable by strong L^r. The positive gradient error on the right converges by the global strong gradient result (3.25). The summed defect is $O(h_j)$. An endpoint shift by at most h_j tends to zero after the L^1 convergence of the right-hand integrands. These facts prove (3.1) at almost every endpoint.

It holds at every endpoint. Along any sequence $t_k \to t$, the uniform H^1, L^{r_1} and moment bounds give global strong L^2 compactness; narrow continuity identifies its limit with U_t. Weak gradient compactness gives $F(U_t) \le \liminf F(U_{t_k})$. Choose t_k from the full-measure endpoint set and use continuity of the indefinite L^1 integrals. This extends (3.1) to all t, including zero.

Near zero its right-hand error is $O(t)$ by the uniform H^1, L^{r_1} bounds. Thus $\limsup_{t\downarrow 0} F(U_t) \le F(\bar{u})$. The preceding compactness and narrow continuity give strong L^2 convergence and the reverse lower semicontinuity, hence $F(U_t) \to F(\bar{u})$. The gradient norms then converge, proving strong H^1 at zero. Interpolation with the uniform L^{r_1} bound gives strong L^r.

Finally set

$$\mathfrak{D}(w) = c_d(\mathcal{C}_\psi(w) + \mathcal{D}_\psi(w)) + \frac{\epsilon}{2}\|\Delta w\|_2^2, \qquad \mathfrak{E}(w) = B_M(w) + 2\chi\epsilon\|\nabla w\|_2^2.$$

The errors $\mathfrak{E}(U_t)$ converge to $\mathfrak{E}(\bar{u})$ at zero. The extended nonnegative functional $\mathfrak{D}$ is lower semicontinuous under this strong H^1 convergence: a finite liminf controls H^2 since $\epsilon > 0$, the weighted Hessian passes weakly, and the quartic term passes by Fatou and almost-everywhere gradients. Rearranging (3.1) gives

$$\frac{F(\bar{u}) - F(U_t)}{t} \ge \frac{1}{t}\int_0^t [\mathfrak{D}(U_s) - \mathfrak{E}(U_s)]\,\mathrm{d}s.$$

Lower semicontinuity and error continuity prove (3.2), also when $\mathfrak{D}(\bar{u}) = +\infty$. This completes the proof of Proposition 3.1.

4 Localized entropy and finite propagation

We return to the original minimizing movements (2.2). All auxiliary steps and periodic boxes used in Section 3 have already been removed.

Proposition 4.1. *The original global weak curve can be selected so that, for each cutoff ζ in any prescribed countable family of smooth functions taking values in $[0,1]$ and constant outside compact sets, and almost every $t>0$,*

$$\int \zeta^4 u(t)^p + c_d \int_0^t \int \zeta^4 [u^a |\nabla^2 u|^2 + u^{a-2} |\nabla u|^4] \\ \le \int \zeta^4 \rho_0^p + C_d \int_0^t \int u^{p+1} [|\nabla \zeta|^4 + \zeta^2 |\nabla^2 \zeta|^2 + \chi^2 \zeta^4]. \tag{4.1}$$

The positive constants depend only on d through the fixed exponent (2.5); they are independent of the cutoff radius, width, time and approximation parameters.

Proof. Use the flow-interchange inequality of [4, Theorem 3.2, (3.21)]. For the actual λ_M-flow of G, with $D^G F(v) = \limsup_{s \downarrow 0} [F(v) - F(S_s v)]/s$, it reads

$$G(u_k) + \tau D^G F(u_k) + \frac{\lambda_M}{2} W_2^2(u_k, u_{k-1}) \le G(u_{k-1}).$$

Every u_k is an H^1 density in $\mathcal{P}_2(\mathbb{R}^d)$ with finite entropy, so all domains match. Proposition 3.1 bounds this dissipation from below. Its possibly infinite positive terms must then be finite at a minimizer, since the displayed upper bound is finite. Summation, dropping the positive ϵH^2 term, gives

$$G(u_N) + c_d \tau \sum_{k=1}^N [\mathcal{C}_\psi(u_k) + \mathcal{D}_\psi(u_k)] \le G(\rho_0) + \tau \sum_{k=1}^N [B_M(u_k) + 2\chi\epsilon \|\nabla u_k\|_2^2] \\ + \frac{|\lambda_M|}{2} \sum_{k=1}^N W_2^2(u_k, u_{k-1}). \tag{4.2}$$

At fixed M, ϵ the final term is $O_M(\tau)$ by (2.4).

First let $\tau \to 0$ along the original global subsequence. Strong spacetime H^1, L^r convergence passes the density and gradient errors. For the positive terms use the bounded continuous weight q_M with weak Hessians, and almost-everywhere gradient convergence with Fatou for the quartic term. At almost every endpoint the lower semicontinuity of G and the common moment bound apply. These bounds also give a uniform lower entropy bound. Letting $\epsilon \to 0$ next therefore removes the entropy and heat error and leaves the weighted f_M inequality for the same original curve.

Finally $M \to \infty$. The functions q_M, f_M increase to $u^a, u^p/(ap)$. The error coefficients in (2.19) converge pointwise to constants times u^a and are dominated using (2.10) by $C_d(u^{2+a} + u^{2+3a})$. Monotone convergence on positive terms and dominated convergence on errors are thus legitimate. Multiplication by ap gives (4.1). Indeed

$$\frac{|\nabla \psi|^4}{\psi^3} = 256 |\nabla \zeta|^4, \qquad \frac{|\Delta \psi|^2}{\psi} \le C_d (|\nabla \zeta|^4 + \zeta^2 |\nabla^2 \zeta|^2),$$

and Young's inequality bounds $\chi |\Delta \psi|$ by the same cutoff terms plus $C_d \chi^2 \zeta^4$. Countable cutoff families, integer horizons, integer M and a sequence $\epsilon \downarrow 0$ suffice, so a common refinement preserves one global original curve and all its source weak-solution properties. □

Fix in Proposition 4.1 the countable family of radial cutoffs with inner radius $s = m + 1 - 2^{-j}$ and width $\delta = 2^{-j-1}$, for integers $m \ge 1$ and $j \ge 0$. Each is zero on B_s, one outside $B_{s+\delta}$, and

satisfies $|\nabla\zeta| \le C_d\delta^{-1}$ and $|\nabla^2\zeta| \le C_d\delta^{-2}$. These bounds are uniform for $s \ge 1$. Only these pairs (s,δ) are needed below.

Fix $T > 0$ and choose $R_0 \ge 1$ containing the essential support of ρ_0. For $s \ge R_0$ let

$$A(s) = \int_0^T \int_{|x|>s} u^{p+1}\,\mathrm{d}x\,\mathrm{d}t. \tag{4.3}$$

It is finite and tends to zero as $s \to \infty$, by the uniform H^1 bound and the strict subcritical exponent. For any of the fixed pairs (s,δ) with $s \ge R_0$, use the corresponding cutoff ζ. The initial term in (4.1) vanishes, giving

$$\operatorname*{ess\,sup}_{0<t<T} \int \zeta^4 u^p + \int_0^T \int \zeta^4[u^a|\nabla^2 u|^2 + u^{a-2}|\nabla u|^4] \le C_d(\delta^{-4} + \chi^2)A(s). \tag{4.4}$$

For the time integral, let admissible endpoints tend to T.

Lemma 4.2. *Put*

$$b = \frac{p+1}{2} = 1 + \frac{a}{2}, \qquad q = \frac{2p}{p+1}, \qquad k = \frac{2(p+1)}{p} > 2, \qquad w = \zeta^k u^b. \tag{4.5}$$

Then $w \in L^2(0,T;H^2(\mathbb{R}^d))$ *and*

$$\operatorname*{ess\,sup}_{t<T} \|w(t)\|_q^q + \int_0^T \|\nabla^2 w(t)\|_2^2\,\mathrm{d}t \le C_d(\delta^{-4} + \chi^2)A(s). \tag{4.6}$$

Proof. The identities $kq = 4$, $bq = p$ give $|w|^q = \zeta^4 u^p$. We supply the second-order chain justification because no unweighted L^4 gradient embedding is available in high dimension. Regularize the power at zero by $\Phi_\eta(z) = (z+\eta)^b - \eta^b$. Multiply its derivative by a smooth upper cutoff $\theta(z/K)$, equal to one below K and zero above $2K$, and integrate from zero, with $K \ge \eta$. The resulting $\Phi_{\eta,K}$ has bounded first and second derivatives, vanishes at zero, and satisfies, with constants independent of η, K,

$$|\Phi'|^2 \le C_d(z^a + \eta^a), \qquad |\Phi''|^2 \le C_d z^{a-2} \quad (z > 0),$$
$$|\Phi(z)|^2 \le C_d(z^{2+a} + \eta^a z^2).$$

On $K < z < 2K$ the added second derivative has the same bound since z is comparable to K. Writing $\sigma = \zeta^k$, the distributional identity is

$$\begin{aligned}\nabla^2[\sigma\Phi(u)] &= \sigma\Phi'(u)\nabla^2 u + \sigma\Phi''(u)\nabla u \otimes \nabla u \\ &\quad + 2\,\mathrm{sym}[\nabla\sigma \otimes \Phi'(u)\nabla u] + \Phi(u)\nabla^2\sigma.\end{aligned} \tag{4.7}$$

Initially this is an L^1_{loc} chain/product identity: the truncated first derivative is bounded in spatial $W^{1,2}_{\mathrm{loc}}$, and its product with $\nabla u \in H^1$ is in $W^{1,1}_{\mathrm{loc}}$. The estimates now show that its right side is L^2.

Since $2k \ge 4$, the first two squared terms are bounded in spacetime by

$$C_d \int_0^T \int \zeta^4[u^a|\nabla^2 u|^2 + u^{a-2}|\nabla u|^4] + C_d\eta^a \int_0^T \int |\nabla^2 u|^2.$$

For the cross term use

$$\zeta^{2k-2}|\nabla\zeta|^2 u^a|\nabla u|^2 \le c\zeta^4 u^{a-2}|\nabla u|^4 + C_{d,c}\zeta^{4k-8}|\nabla\zeta|^4 u^{a+2}.$$

Here $4k-8>0$. The η^a part is bounded by a vanishing multiple of $\int_0^T\|\nabla u\|_2^2$. Finally

$$|\nabla^2\sigma|^2 \le C_d[\zeta^{2k-4}|\nabla\zeta|^4+\zeta^{2k-2}|\nabla^2\zeta|^2].$$

This is supported in the transition annulus and bounded by $C_d\delta^{-4}$. Its contribution is at most $C_d\delta^{-4}A(s)$ plus a vanishing η^a multiple of $\int_0^T\|u\|_2^2$. First let $K\to\infty$ and then $\eta\downarrow 0$. The powers converge in spacetime L^2 and the Hessians are weakly compact in L^2; all η^a terms vanish by the original H^1, H^2 bounds. Thus the Hessian estimate passes to w. Global L^2 and Hessian L^2 imply inhomogeneous H^2 by derivative interpolation. Combining with (4.4) proves (4.6). □

Proof of Theorem 1.1. For the exponent q in (4.5), Lemma A.1 gives

$$\|w\|_2 \le C_d\|\nabla^2 w\|_2^{\theta}\|w\|_q^{1-\theta}, \qquad \theta=\frac{d}{d+4p}.$$

Integrate its square in time, use Hölder and (4.6), and note that $w=u^b$ outside $B_{s+\delta}$. We obtain

$$A(s+\delta)\le C_dT^{\gamma}(\delta^{-4}+\chi^2)^{\beta}A(s)^{\beta}, \qquad \gamma=\frac{4p}{d+4p}, \qquad \beta=1+\frac{4}{d+4p}>1. \tag{4.8}$$

Indeed $\gamma=1-\theta$ and $\theta+2(1-\theta)/q=\beta$.

To see exactly how this forces a zero tail, choose an integer $s_*\ge R_0$ and put $s_j=s_*+1-2^{-j}$, $\delta_j=2^{-j-1}$, $y_j=A(s_j)$ for $j\ge 0$. Equation (4.8) implies

$$y_{j+1}\le KB^j y_j^{\beta}, \qquad K=C_dT^{\gamma}(1+\chi^2)^{\beta}2^{4\beta}, \quad B=2^{4\beta}.$$

If $y_0\le K^{-1/(\beta-1)}B^{-1/(\beta-1)^2}$, induction gives

$$y_j\le K^{-1/(\beta-1)}B^{-j/(\beta-1)-1/(\beta-1)^2}\longrightarrow 0.$$

Such a finite integer s_* exists because $A(s)\to 0$. Monotonicity then gives $A(s_*+1)=0$. Nonnegativity implies that $u=0$ almost everywhere on $(0,T)\times(\mathbb{R}^d\setminus B_{s_*+1})$.

This uses only the countable family fixed before the tail estimate. Taking integer time horizons in the preceding argument gives the support property for the same original global curve on all those horizons; any finite T is contained in a larger integer horizon. Its narrow continuity, probability and moment properties, regularity and exact weak identity are the unchanged CEFF conclusions from Section 2. This proves every clause of Theorem 1.1. □

A Two interpolation estimates

Lemma A.1. *If $1\le q<2$ and $w\in H^2(\mathbb{R}^d)\cap L^q(\mathbb{R}^d)$, then*

$$\|w\|_2\le C_{d,q}\|\nabla^2 w\|_2^{\theta}\|w\|_q^{1-\theta}, \qquad \theta=\frac{d(1/q-1/2)}{2+d(1/q-1/2)}.$$

Proof. Choose an even nonnegative smooth unit kernel η_h and put $s=d(1/q-1/2)$, $L=\|w\|_q$, $D=\|\nabla^2 w\|_2$. Convolution is L^q-contractive and $\|\eta_h*w\|_\infty\le Ch^{-d/q}L$. Interpolating these two bounds gives $\|\eta_h*w\|_2\le Ch^{-s}L$. The first moment of the kernel is zero, so Taylor's integral remainder and translation continuity in L^2 give $\|w-\eta_h*w\|_2\le Ch^2D$. This is first proved for smooth w and passed by H^2 approximation. Thus $\|w\|_2\le C(h^{-s}L+h^2D)$ for every $h>0$. If either norm is zero, the conclusion follows directly, letting $h\to\infty$ when $D=0$. Otherwise choose $h=(L/D)^{1/(s+2)}$. This proves the formula; for $q=2p/(p+1)$ it gives $\theta=d/(d+4p)$ in every finite dimension. □

Lemma A.2. *For a nonnegative periodic $H^1(Q_L)$ density of mass one and $L \geq 1$,*

$$\frac{1}{2}\|\nabla u\|_2^2 - \chi\|u\|_2^2 \geq \frac{1}{4}\|\nabla u\|_2^2 - C_{d,\chi}.$$

Proof. Let $c_n = (2L)^{-d}\int_{Q_L} u(x)e^{-i\pi n\cdot x/L}\,\mathrm{d}x$. Mass one gives $|c_n| \leq (2L)^{-d}$. Splitting Parseval's sum at physical frequency $K > 0$, a lattice count and the gradient norm give

$$\|u\|_2^2 \leq C_d(K^d + L^{-d}) + K^{-2}\|\nabla u\|_2^2.$$

With $E = \|\nabla u\|_2^2 > 0$ choose $K = E^{1/(d+2)}$ to obtain $\|u\|_2^2 \leq C_d E^{d/(d+2)} + C_d L^{-d}$. The constant-density case $E = 0$ has the same bound. Since $d/(d+2) < 1$, scalar Young's inequality proves the assertion uniformly for $L \geq 1$. □

AI-use statement

This manuscript was developed with substantial AI assistance in mathematical exploration, proof drafting, source checking and exposition. Two fresh AI contexts provided reader and referee-style feedback on a frozen draft, followed by author revision. These AI checks do not constitute independent human review or formal verification. Independent mathematical assessment of the argument is requested.

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